MPAS-URBAN QUICKSTART | GUIDE 3 OF 3

Run the HRAIN2025 case

Prepare the run directory, apply the 500 m mesh setting, and optionally use the HRAIN2025 NTDK cutoff modification.

OUTCOME You will launch the three-day HRAIN2025 atmosphere experiment and understand which settings are standard, case-specific, or required only for comparison with existing historical results.

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Before you begin

Complete the recommended official MPAS tutorial and Guides 1–2. Confirm that the initial-condition file and hk_hull_500m_graph.info.part.112 exist.

Choose the executable deliberately: standard hkust-dev for normal MPAS-Urban learning and new experiments, or the verified HRAIN2025 case-specific commit 7c1610b8f41781c2433f780bd7496da63b25a920 for the optional 10 km CU cutoff and HRAIN2025 ZT compatibility setting. Soil categories are already stored in the static file; use the static and initial-condition files created in Guide 2 for the chosen soil dataset.

Step 1 | Prepare the run directory

Guide 2 created an independent case under $HOME/MPAS. MPAS_BUILD is the tutorial-defined path to the compiled MPAS source tree, not an official MPAS variable. Set it again when using a new shell, select the same build used for initialization, and link all runtime files from that build into the existing case.

SELECT THE SAME COMPILED BUILD

```
export MPAS_ROOT="$HOME/MPAS"
export MPAS_BUILD="$MPAS_ROOT/HKUST-MPAS"

test -x "$MPAS_BUILD/atmosphere_model"
```

For the verified HRAIN2025 comparison/cutoff build, replace MPAS_BUILD with $MPAS_ROOT/HKUST-MPAS-NTDK. Do not mix files from the two builds.

For a fully comparable HRAIN2025 simulation, continue from a static file generated with the Dy and Fung (2016) dataset and the corresponding initialization setup. A static file created with the official MPAS BNU data uses publicly downloadable soil data that are similar, but not identical, to the HRAIN2025 soil dataset. Use separate case directories for each soil/source combination.

ENTER THE EXISTING CASE

```
export HRAIN_REPO="$MPAS_ROOT/MPAS-Urban-HRAIN2025"
export CASE="$MPAS_ROOT/HRAIN2025_30km_500m"

test -d "$CASE"
cd "$CASE"
test -s 30km_500m.init.nc
test -s hk_hull_500m_graph.info.part.112
```

COPY THE RUN CONFIGURATION

```
cp "$HRAIN_REPO/config/namelist.atmosphere" ./namelist.atmosphere
cp "$MPAS_BUILD/streams.atmosphere" ./streams.atmosphere
```

LINK ALL RUNTIME FILES FROM THE SELECTED BUILD

```
ln -sf "$MPAS_BUILD/atmosphere_model" ./atmosphere_model
ln -sf "$MPAS_BUILD"/default_inputs/stream_list.atmosphere.* .
ln -sf "$MPAS_BUILD"/src/core_atmosphere/physics/physics_wrf/files/*TBL .
ln -sf "$MPAS_BUILD"/src/core_atmosphere/physics/physics_wrf/files/*DATA* .
ln -sf "$MPAS_BUILD"/src/core_atmosphere/physics/physics_noahmp/parameters/*TBL .
```

FINAL FILE CHECK

```
ls -lh atmosphere_model
ls -lh namelist.atmosphere streams.atmosphere
ls -lh 30km_500m.init.nc hk_hull_500m_graph.info.part.112
```

BUILD CONSISTENCY NOTE The executable, streams file, stream lists, physics tables, and data links must all come from the same MPAS_BUILD. Do not relink a case after it has produced output; use a separate case directory when comparing source versions.

Step 2 | Set the case time and mesh scale

REFERENCE NAMELIST https://github.com/Liuzh223/MPAS-Urban-HRAIN2025/blob/main/config/namelist.atmosphere

FILE TO EDIT namelist.atmosphere. Copy the repository version into the run directory and verify the following fields before launch.

CHANGE 1 - config_start_time: the first model time is 2025-08-03_00:00:00.

CHANGE 2 - config_run_duration: 3_00:00:00 covers the complete three-day case; the commented three-hour value is only a smoke test.

CHANGE 3 - config_dt: 6 seconds is the dynamics time step used by this experiment.

CHANGE 4 - config_len_disp: explicitly set 500.0 here in the running namelist. This is the first point in the tutorial where this case parameter is introduced.

NAMELIST.ATMOSPHERE

```
&nhyd_model
    config_start_time   = '2025-08-03_00:00:00'
    config_run_duration = '3_00:00:00'
    config_dt           = 6
    config_len_disp     = 500.0
/
```

REQUIRED CASE SETTING Keep config_len_disp = 500.0. It explicitly records the 500 m minimum mesh spacing used by this experiment and should not be omitted even when the mesh contains nominalMinDc.

Step 3 | Set the physics schemes and optional cutoff

FILE TO EDIT Continue editing the &physics group in namelist.atmosphere. The following suite is the explicit HRAIN2025 case configuration; verify every value rather than relying on source-tree defaults.

Physics suite used by HRAIN2025

The case uses WSM6 microphysics, New Tiedtke cumulus, Noah-MP land surface, YSU boundary layer and gravity-wave-drag options, cloud fraction, RRTMG longwave and shortwave radiation, the revised Monin-Obukhov surface layer, and MPAS-Urban. Use the same suite with either source tree; the standard hkust-dev build omits config_cu_disable_dx.

```
&physics
    config_microp_scheme       = 'mp_wsm6'
    config_convection_scheme   = 'cu_ntiedtke'
    config_lsm_scheme          = 'sf_noahmp'
    config_pbl_scheme          = 'bl_ysu'
    config_gwdo_scheme         = 'bl_ysu_gwdo'
    config_radt_cld_scheme     = 'cld_fraction'
    config_radt_lw_scheme      = 'rrtmg_lw'
    config_radt_sw_scheme      = 'rrtmg_sw'
    config_sfclayer_scheme     = 'sf_monin_obukhov_rev'
    config_urban_physics       = true
/
```

Optional HRAIN2025 cutoff derivative

Use the following only when atmosphere_model was compiled from the verified HRAIN2025 case-specific commit 7c1610b8f41781c2433f780bd7496da63b25a920 on Liuzh223/HKUST-MPAS-LIU branch ntdk-grid-cutoff.

```
&physics
    config_convection_scheme = 'cu_ntiedtke'
    config_cu_disable_dx     = 10000.0  ! m; set 0.0 to disable cutoff
/
```

The value 10000.0 is a 10 km grid-spacing threshold. With config_convection_scheme = 'cu_ntiedtke', New Tiedtke convection is suppressed where local mesh spacing is below 10 km. Set config_cu_disable_dx = 0.0 to disable this cutoff; the New Tiedtke scheme then remains active according to its normal configuration. Non-positive values do not activate the threshold.

SOURCE NOTE ntdk-grid-cutoff is a case-author derivative, not the official HKUST-MPAS main branch. The verified case commit is https://github.com/Liuzh223/HKUST-MPAS-LIU/commit/7c1610b8f41781c2433f780bd7496da63b25a920. There is no separate shell command named cutoff.

Step 4 | Check the decomposition and streams

CHANGE 1 - namelist.atmosphere/&decomposition: use the prefix hk_hull_500m_graph.info.part. and launch 112 MPI tasks.

```
&decomposition
    config_block_decomp_file_prefix =
      'hk_hull_500m_graph.info.part.'
/
```

Use 112 MPI tasks so that the prefix resolves to hk_hull_500m_graph.info.part.112. In streams.atmosphere, point the input stream to the initial-condition file created in Guide 2 and select the desired output interval.

FILE TO EDIT streams.atmosphere. Set the input filename to 30km_500m.init.nc, choose the history filename and output interval, and verify that all relative paths are resolved from the run directory.

Step 5 | Run and monitor

RUN

```
cd "$CASE"
export OMP_NUM_THREADS=1

mpiexec -n 112 ./atmosphere_model
```

CHECK

```
tail -n 60 log.atmosphere.0000.out
ls -lh history.*.nc 2>/dev/null || ls -lh *.nc
```

For an inexpensive first check, temporarily use a short config_run_duration, confirm that output is created, and then restore the intended three-day duration. Do not describe the short smoke test as the HRAIN2025 experiment result.

Final note | Comparing with existing HRAIN2025 results

The HRAIN2025 historical experiment results have not yet been publicly released. This section is needed only for a future direct comparison with those existing results. It is not required for learning MPAS or for a new independent simulation.

Use the verified HRAIN2025 case-specific source commit: https://github.com/Liuzh223/HKUST-MPAS-LIU/commit/7c1610b8f41781c2433f780bd7496da63b25a920. This v8.2.2-based source contains the 10 km CU cutoff and the HRAIN2025 ZT compatibility setting, and adds config_soilcat_data from MPAS v8.3+; the rainfall branch is not required.

SOIL DATA HRAIN2025 static interpolation used the updated global WRF soil map described by Dy and Fung (2016). Readers needing a fully comparable HRAIN2025 simulation should consult the paper and contact its authors for the same data and processing details. The official MPAS BNU data are publicly downloadable and similar, but not identical, to that dataset, so a run using a BNU static file is not an exact reproduction.

SOURCE FILE

```
src/core_atmosphere/physics/physics_wrf/module_sf_urban.F
```

Inside SFCDIF_URB, the case-specific source has two active ZT updates: one after USTAR is calculated and one inside the stability iteration. The behavior originates from upstream HKUST-MPAS commit 562bdb4e5ef63527496b6374d52def06a01a84a0: https://github.com/HKUST-MPAS/HKUST-MPAS/commit/562bdb4e5ef63527496b6374d52def06a01a84a0. Z0 is the momentum roughness length and ZT is the thermal roughness length; this setting does not redefine or recalculate Z0.

```
ZT = 0.1 * Z0
```

The verified case-specific commit already includes both active ZT = 0.1 * Z0 assignments, so no manual source edit is needed. Check out the exact commit and rebuild atmosphere_model. Do not confuse this with Z0HC_TBL = 0.1 * Z0C_TBL, which is a different calculation.

For a new independent experiment, keep the ZT implementation supplied by the selected HKUST-MPAS version and record the repository, branch, and commit. Do not modify ZT merely to complete the tutorial.

Completion checklist

- config_start_time is 2025-08-03_00:00:00.
- config_run_duration is 3_00:00:00 for the full experiment.
- config_len_disp is explicitly 500.0.
- config_cu_disable_dx is present only for a verified cutoff-capable build; it is 10000.0 for this case, while 0.0 disables the cutoff.
- The MPI task count matches .part.112.
- For direct comparison with the original HRAIN2025 configuration, use the exact case-specific atmosphere commit and the Dy and Fung (2016) static-input dataset. A static file created with the official MPAS BNU data uses publicly downloadable soil data that are similar, but not identical, to the HRAIN2025 soil dataset.

Primary references

- Dy and Fung (2016), updated global WRF soil map
- Official MPAS-A tutorial
- MPAS-A Quick Start Guide
- HKUST-MPAS hkust-dev
- HRAIN2025